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PAPER_983: SCm First Axiom Validation — $|U_b/U_g| > 0.5$ at $25.4 \mu\text{m}$

Abstract

The SCm First Axiom states that at the characteristic wavelength $\lambda_{\text{SCm}} = 25.4 \mu\text{m}$ (corresponding to $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$), the buoyancy-to-gravity ratio must exceed 0.5, i.e., $|U_b/U_g| > 0.5$. This paper validates the axiom numerically for the Solar System ($M_\odot$, $r = 1 \text{ AU}$) and demonstrates that the measured ratio $|U_b/U_g| \approx 1.536$ satisfies the threshold with margin, confirming the SCm vacuum is buoyancy-active at its fundamental resonance.

1. First Axiom Statement

$$\text{Axiom 1: } \left| \frac{U_b(r, \lambda_{\text{SCm}})}{U_g(r)} \right| > 0.5$$

This ensures the SCm vacuum exerts non-negligible buoyancy at its own characteristic scale — a self-consistency requirement for any medium that claims to support buoyancy forces.

2. Ratio Computation

$$\frac{|U_b|}{|U_g|} = \frac{\sum_{i=1}^{26} \frac{GM}{r^2} e^{-[\text{SSq}] \cdot i/26} \cdot \beta_i}{\sum_{i=1}^{26} \frac{GM}{r^2} \cdot [\text{SSq}] \cdot i/26}$$

The GM/r^2 factors cancel:

$$= \frac{\beta_i \sum_{i=1}^{26} e^{-[\text{SSq}] \cdot i/26}}{[\text{SSq}] \sum_{i=1}^{26} i/26} = \frac{\beta_i \cdot S_{26,\text{exp}}}{[\text{SSq}] \cdot 13.5}$$

3. Numerical Result

With $[\text{SSq}] = 0.57$, $\beta_i = 0.603$:

$$\frac{|U_b|}{|U_g|} = 1.536 > 0.5 \quad \checkmark$$

The ratio exceeds the threshold by a factor of ~ 3 , indicating strong buoyancy activation.

4. Physical Interpretation

- $|U_b/U_g| > 1$: buoyancy dominates gravity $\rightarrow$ net outward force
- This is consistent with the heliospheric expansion observed at $r = 1$ AU
- The axiom would fail for $\beta_i < 0.196$, setting a lower bound on buoyancy coupling

5. Implementation

Class `SCmFirstAxiomValidator` in `fubi_master_calculator.py`: computes $|U_b/U_g|$ at $\lambda = 25.4 \mu\text{m}$, asserts ratio > 0.5 .

References

- PAPER_979: Complete 6-Layer F_U_Bi_i
 - PAPER_980: Solar Surface Calibration
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§A. Cosmogenesis-Linked Lagrangian

The First Axiom constrains the Lagrangian: $\partial V_b/\partial\phi > 0.5 \cdot \partial V_g/\partial\phi$ at λ_{SCm} , ensuring the buoyancy sector is dynamically relevant.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS**: The axiom sets a minimum vacuum density for buoyancy activation.
- **DVP**: Dipole alignment at $25.4 \mu\text{m}$ implies vortex coherence over $\sim 10^4$ lattice sites.
- **BSH**: The harmonic sum $S_{26,\text{exp}}$ determines the axiom threshold through its convergence properties.